GAS PROCESSING & METHANOL COMPLEX PROJECT NO.11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**HAZID Terms of Reference**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-012
Document Revision	R01
Document Status	Issued for Review
Originator	Ekene Okoye
Security Classification	Restricted
Issue Date	6-Jul-22

HAZID Terms of Reference

R01	06.07.22	Issued for Review	E. Okoye	E. Maduabuchi	K. Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	06.07.22	Issued for Review
R02		
A01		

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
   for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-012

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2.The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
ALARP	As Low As Reasonably Practicable
DEP	Design and Engineering Practice
FEED	Front End Engineering Design
HAZID	Hazard Identification
MAH	Major Accident Hazards
PEFS	Process Engineering Flowscheme
PFS	Process Flowscheme
RAM	Risk Assessment Matrix

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity Greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35&36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: BubouweBou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

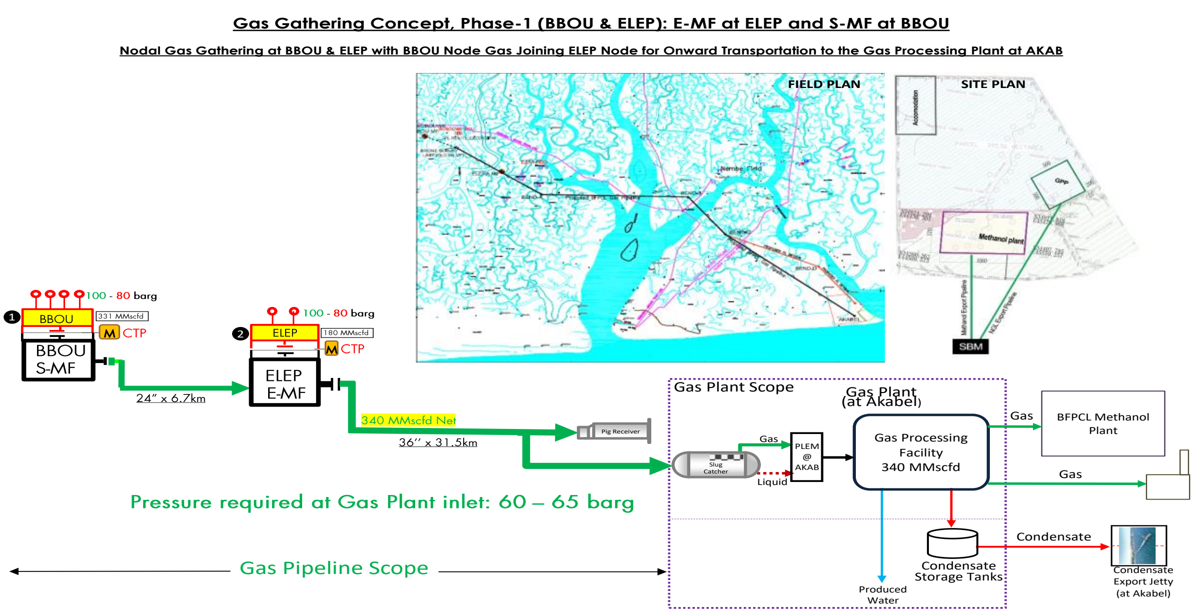

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The objective of the HAZID (Hazard Identification) is to identify the hazards applicable to the project in the FEED phase and carry out a risk assessment rating, address the control and recovery preparedness measures and make recommendations as appropriate

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Methodology

The study method will be a combination of identification, analysis, and brainstorming on the hazards identified within the project by breaking down the project into various nodes and "following the molecules". The identified hazards will be assigned a risk ranking. Risk mitigation plans will be identified and recommendations will be made if the controls are not considered sufficient

The output of the HAZID Study will be a record of the discussion detailing the identified hazards, Project Hazard and Effect Register, and recommendations for any risk mitigation actions required for the work covered under the scope

The basic approach will involve:

- Assembly of an appropriate team; design team engineers, construction engineers, HSE reps, asset team representatives, in a workshop
- Short presentations by the project team detailing the designs and systems, and construction activities under the scope of the HAZID Study
- Identify hazards related to the "molecules"
- Each of the hazards identified was ranked for severity, the likelihood of occurrence, and consequences on people, assets, environment, and community using the Risk Assessment Matrix (Appendix 2).
- Carry out hazard analysis for all the hazards /scenarios and bowtie analysis for the major accident hazards (MAHs)
- Record the discussions on worksheets (Hazard & Effect Register) summarizing the nature of the hazard, its consequences, the current safeguards in place, and recommendations for any actions required.

3.1. Guidewords

HAZID guidewords applicable to the phase of the project (FEED) shall be adopted for the review session. The guidewords are intended to prompt discussion and are not intended to be exhaustive.

3.2. Preparation Work, Nodes, and Documentation

A preliminary note delineation has been carried out by the Facilitator (ESLP Consultants Limited). The proposed nodes of study are in Table 1 below:

Table 1: Proposed HAZID Nodes

Node	Title
1	BBOU Manifold
2	ELEPA Manifold
3	AKABEL Manifold

3.3. Preparation Work, Nodes, and Documentation

The following key documentation will be required for the HAZID study

- Facility layout/plot plan incl. major equipment layout and occupied buildings
- Emergency response plans
- Plans for construction, transportation, installation and commissioning
- Neighbouring facilities
- Approved for HAZOP Process Flow Schemes
- Approved for HAZOP Process Engineering Flowschemes
- Risk Assessment Matrix (RAM)

3.4. HAZID Agenda

The agenda will be as follows:

1. Team introduction, domestic arrangements, safety instructions by Client
2. HAZID Chairman (Facilitator) introduce the scope of the HAZID study
3. Project Team presents the scope of the project, execution strategies, highlighting the key features and decisions taken to date
4. The HAZID Chairman will explain the HAZID process including the HAZID worksheet format and terminologies
5. The Chairman will:
    i. Propose Nodes and agree on Nodes with the team.
    ii. Commence HAZID using the guidewords. This is not a rigid list and the team can add additional hazards not covered by these guidewords as required.
6. All relevant information will be recorded on the worksheet.
7. Refreshments & comfort breaks shall be held at intervals as agreed by the team

3.5. HAZID Session Details

The HAZID will be carried out on the 7th of July 2022. The session will be performed for 1 day (estimate) commencing at 09.00AM and finishing by 5:00PM. Attendees/reps are shown in

Table 2 below:

Table 2: HAZID Attendee list

Discipline	Name	Company
ESLP Consultants Reps		
HAZID Chairman	Engr. Emeka Maduabuchi	ESLP Consultants Ltd
HAZID Scribe	Okoye Ekene m.	ESLP Consultants Ltd

The HAZID shall be recorded in Excel worksheet. The worksheets shall be projected onto a screen so that team members have an opportunity to review and comment during the session. Actions recorded during the session shall be assigned to individual team members for follow-up and close-out after the meeting.

3.6. Deliverables & Follow Up Work

A HAZID worksheet presenting a record of the HAZID findings will be issued to all attendees for review to confirm that the report reflects the session content. A HAZID report will be issued to the HAZID Co-ordinator, with the HAZID worksheet as an attachment to the report.

HAZID action tracking sheet will be maintained by the Project Team HSE reps.

Appendix 1: Typical HAZID Guidewords

Section 1: External & Environmental Hazards				
S/N	Category	Hazard/Event	Scenario	Y/N
1.01	Natural and Environmental Hazards	Climate Extremes	Temperature, waves, wind, dust, flooding, sandstorms, ice, blizzards	
1.02		Lightening		
1.03		Earthquakes		
1.04		Erosion	Ground slide, coastal, riverine	
1.05		Subsidence	Ground structure, foundations	
2.01	Created (Man-made) Hazards	Security Hazards	Internal and external security threats	
2.02		Terrorist Activity	Riots, civil disturbance, strikes, military action, political unrest	
3.01	Effect of the Facility on the Surroundings	Geographical – Infrastructure	Plant location, plant layout, pipeline routing	
3.02		Proximity to Population	Proximity to population	
3.03		Adjacent Land Use	Crop burning, airfields, accommodation camps	
3.04		Proximity to Transport Corridors	Shipping lanes, air routes, roads, etc	
4.01	Environmental Damage	Continuous Plant Discharges to Air	Flares, vents	
4.02		Continuous Plant Discharges to Water	Target/legislative requirements, drainage facilities	
4.03		Continuous Plant Discharges to Soil	Drainage, chemical storage	
4.04		Emergency/upset Discharges	Flares, vents, drainage	
Section 2: Facility Hazards				

Code	Category	Hazard/Event	Scenario	
5.01		Maintenance Philosophy	Heavy lifting, access	
5.02		Emergency Response	Isolation, ESD philosophy, blowdown, flaring/venting requirements	
5.03		Concurrent Operations	Production, maintenance requirements	
5.04		Start-up / Shutdown		
6.01	Fire and Explosion Hazards	Sources of Ignition	Electricity, flares, sparks, hot surfaces (mitigation measures include: identify, remove, separate)	
6.02		Equipment Layout	Confinement, escalation following release of explosive or flammable fluid; Mitigation measures include reduce degree of confinement, spacing based on consequence assessment, escalation barriers)	
6.03		Fire Protection and Response	Active/passive insulation, fire/gas detection, blowdown/relief system philosophy, firefighting facilities	
6.04		Operator Protection	Means of escape, PPE, communications, emergency response, plant evacuation	
7.01	Process Hazards	Asset Integrity Management	Process control failure, structural failure, erosion or corrosion (mitigation measures include: recognise and mini-	

			mise process hazards during design, inherently safe plant, containment and recovery measures)	
7.02		Excess/zero Level	Overfill storage tanks and vessels, blowby to downstream vessels	
8.01	Utility Systems	Firewater Systems		
8.02		Fuel Gas		
8.03		Heating / Cooling Medium		
8.04		Potable Water		
9.01	Maintenance Hazards	Access Requirements		
9.02		Heavy Lifting Requirements		
9.03		Transport		
10.01	Construction/ Existing Facilities	Tie-ins (shutdown requirements)		
10.02		Concurrent Operations		
10.03		Skid Dimensions (weight handling/equipment (congestion)		
Section 3: Others				
11.00	Biological	snakes, reptiles		
12.00	Chemicals	glycol, foam systems, water treatment chemicals		
13.00	Ergonomics	Human machine interface		
14.00	Noise	Blowdown valves, gas generator		

Appendix 2: Risk Assessment Matrix

SEVERITY	CONSEQUENCES				INCREASING LIKELIHOOD				
					A	B	C	D	E
	People	Assets	Environment	Reputation	Never heard of in the Industry	Heard of in the Industry	Has happened in the Organisation or more than once per year in the Industry	Has happened at the Location or more than once per year in the Organisation	Has happened more than once per year at the Location
0	No injury or health effect	No damage	No effect	No impact					
1	Slight injury or health effect	Slight damage	Slight effect	Slight impact					
2	Minor injury or health effect	Minor damage	Minor effect	Minor impact					
3	Major injury or health effect	Moderate damage	Moderate effect	Moderate impact					
4	PTD or up to 3 fatalities	Major damage	Major effect	Major impact					
5	More than 3 fatalities	Massive damage	Massive effect	Massive impact					